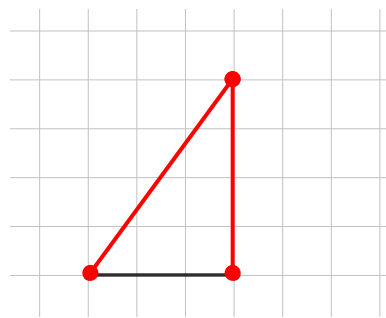
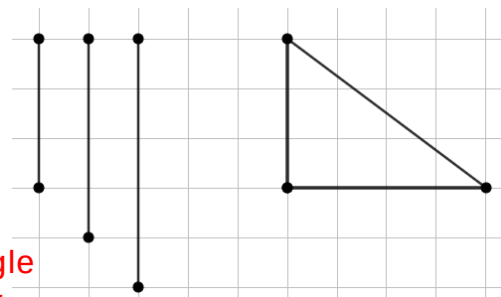


Three line segments and a triangle are shown.

1. Compare the triangle that was constructed with the three line segments. Describe any similarities or differences.

A line segment and one side of the triangle measures 3 units, a line and one side of the triangle measure 4 units. Patty paper can be used to show the third side matches the 5 unit segment.

2. One of the line segments is shown on the grid. Draw a different triangle that uses the other two line segments.



3. Describe how you could determine if the angles of the original triangle and the angles of triangle you drew in question 2 are congruent.

The angles are congruent because all three corresponding sides of the two triangles are congruent. (side-side-side congruence postulate)

4. Complete the statement.

- ☐ Any
☐ Some
☐ None

triangles created using the three line segments are congruent to

the original triangle.

Six triangles are shown.

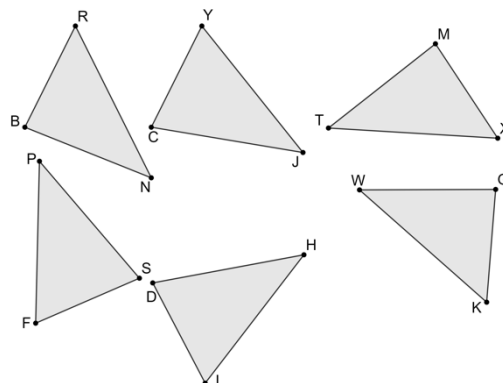
Use patty paper, a blank sheet of paper, a ruler, and/or a protractor to help answer the following questions.

5. Which triangles have a side that has the same length as $\overline{BR}$?

All of them.

6. Which triangles have a side that has the same length as $\overline{BN}$?

$\triangle KGW, \triangle TMX, \triangle BNR$



7. Which triangles have a side that has the same length as $\overline{RN}$?

All of them.

8. Complete the statements.

All of the triangles that have a side that have the same length as $\overline{BR}$

☐ are
☒ are not

congruent to $\triangle RBN$. All of the triangles that have two

sides that have the same length as $\overline{BR}$ and $\overline{BN}$

☐ are
☒ are not

congruent to $\triangle RBN$. All of triangles that have three sides that have

the same length as $\overline{BR}$, $\overline{BN}$, and $\overline{RN}$

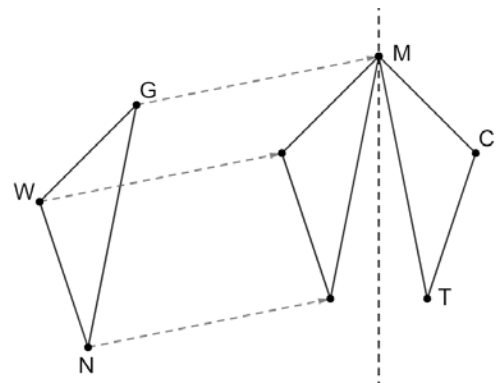
☒ are
☐ are not

congruent to $\triangle RBN$.

A series of transformations shows how $\overline{GW}$, $\overline{GN}$, and $\overline{NW}$ were translated then reflected across a line.

9. Select the reason the transformations that were used ensure the length of each line segment was preserved.

- ☐ The transformations included a dilation.
☒ The transformations were rigid motions.
☒ The transformations did not include a rotation.



10. Complete the statements.

The definition of congruence in terms of rigid motions

ensures that the angles are congruent. Because all of the

parts of $\triangle GNW$ are congruent to all of the corresponding

parts of $\triangle MTC$, the two triangles are congruent. This shows that

the side-side-side congruence postulate is true.